

Author: Now I am going to prove the following converse.

Theorem: If a number b is the limit of a function $f(x)$ at a point a in terms of definition 2, it is also the limit of the function $f(x)$ at a in terms of definition 1.

Proof: In this case I shall use the proof by contradiction. Assume the contrary to what is to be proved, namely, assume that b the limit of $f(x)$ in terms of definition 2, is not, however the limit of $f(x)$ in terms of definition 1. Can you formulate the last proposition (more exactly, the assumption)?

Reader: As far as I remember, a similar formulation has already been discussed in the previous dialogue. If b is not the limit of the function $f(x)$ at point a (in terms of definition 1), it means that there is $\varepsilon' > 0$ such that it is impossible to find a necessary $\delta > 0$. Namely, no matter what δ we select, each time the function $f(x)$ assumes a value outside of $[b - \varepsilon', b + \varepsilon']$ for at least one point x from the δ -neighbourhood of point a , i.e. the inequality $|f(x) - b| < \varepsilon'$ is violated.

Author: Correct. Assume that we have selected precisely this $\varepsilon' > 0$. Next take an arbitrary $\delta > 0$, for instance, $\delta_1 = 1$. As you have said, in any δ -nb of point a and, hence, in the δ_1 -neighbourhood of this point there is at least one point x (denoted by x_1) such that $|f(x_1) - b| > \varepsilon'$.

Reader: What happens if the δ_1 -neighbourhood contains many such points x ?

Author: It makes no difference. The important fact is that there is at least one such point. If there are several such points, take any one of them and denote it by x_1 .

Now we take a new δ , for instance, $\delta_2 = \frac{1}{2}$. According to our assumption, the δ_2 -neighbourhood of point a will contain at least one point x (denoted by x_2) such that $|f(x_2) - b| > \varepsilon'$.

Further we take $\delta_3 = \frac{1}{3}$. The δ_3 -neighbourhood of point a will also contain at least one point x (point x_3) such that $|f(x_3) - b| > \varepsilon'$.